0.1 The Backward Map

The problem is framed as an inverse problem. The backward map takes position and orientation data and calculated 8 flux values from each of the coils. Our goal is to approximate an inverse to this map. To do so, we first try to understand the backward map

0.1.1 Definition and properties.

First we define a pose vector $\mathbf{p} \in \mathbb{R}^5$

$$\mathbf{p} = \begin{bmatrix} x \\ y \\ z \\ \theta \\ \phi \end{bmatrix} \tag{1}$$

In actuality, $\mathbf{p}$ is restricted to a set $\Omega \subset \mathbb{R}^5$

$$\Omega = \left\{ \begin{bmatrix} x \\ y \\ z \\ \theta \\ \phi \end{bmatrix} \in \mathbb{R}^5 : \begin{matrix} x \in (-0.125, 0.125) \\ y \in (-0.125, 0.125) \\ z \in (0, 0.25) \\ \theta \in (0, \pi/2) \\ \phi \in [0, 2\pi) \end{matrix} \right\} \tag{2}$$

Then we define a vector of fluxes, $\mathbf{\Phi} \in \mathbb{R}^8$

$$\mathbf{\Phi} = \begin{bmatrix} \Phi_1 \\ \Phi_2 \\ \Phi_3 \\ \Phi_4 \\ \Phi_5 \\ \Phi_6 \\ \Phi_7 \\ \Phi_8 \end{bmatrix} \tag{3}$$

Now we can write the backward map: $\mathbf{\Phi}(\mathbf{p}) : \mathbb{R}^5 \to \mathbb{R}^8$. More precisely,

$$\Phi_i = \mathbf{H}_i(x, y, z) \cdot \hat{n}(\theta, \phi) \tag{4}$$

where $\mathbf{H}_i$ is the magnetic field calculated at (x, y, z) due to coil i using eqaution (**??**), and $\hat{n}(\theta, \phi)$ is the vector normal to the sensor.

Since this maps from $\mathbb{R}^5$ to $\mathbb{R}^8$ it is clear that the backwards function is not surjective. Looking at injectivity now: we see that

$$\Phi\left(\begin{bmatrix}x\\y\\z\\0\\\phi_1\end{bmatrix}\right)=\Phi\left(\begin{bmatrix}x\\y\\z\\0\\\phi_2\end{bmatrix}\right),\ \forall\phi_1,\phi_2\in\mathbb{R} \tag{5}$$

Therefore this map is not injective either.

0.1.2 The Jacobian

Now that we have established that $\mathbf{\Phi}$ is neither injective nor surjective we must move our attention to thinking about local invertibility rather than global. To do so we define the Jacobian:

$$J(\mathbf{p})=\frac{\partial\mathbf{\Phi}}{\partial\mathbf{p}}\in\mathbb{R}^{8\times 5},\ J_{i,j}=\frac{\partial\Phi_i}{\partial p_i} \tag{6}$$

Since $\mathbf{H}$ is a finite sum of Biot Savart applications it is a smooth map away from the current filaments. $\hat{n}$ is also a smooth map, therefore Φ will also be smooth away from the current filaments and the Jacobian will be well defined there.

A closed form for $\mathbf{J}$ is infesible to find so we instead use finite differences to approximate $\mathbf{J}$

$$\mathbf{J_{:,i}}\approx\frac{\mathbf{\Phi}\left(\mathbf{p}+h*\hat{e}_i\right)-\mathbf{\Phi}(\mathbf{p})}{h} \tag{7}$$

where $\hat{e}_i$ is the unit vector in the ith direction, and h is a small real value ($h=10^{-5}$ in our simulations).